Where should the planner route large Koblitz curves?

2026-10-26 · `research/ic_koblitz_descent_routing_20261026` · protocol frozen in `cd3af7078` · three ecbench sessions, 288 records, all verified, audits OK · reference `rho.signed_frobenius_strong`.

`ic.koblitz_descent` (new in `b561ebca8`) is the ledger §23 one-target Koblitz pipeline, charged in counted group operations. On the same seeds its logs and native counts are identical to `ic price --single-target`. It ran in one session per curve against the planner's `ic.pipeline` recipe and the strong rho.

curve	$\log_2 r$	S pipeline	S descent	descent / pipeline	pipeline / rho	descent / rho
`icv1-f2m47-t22705043-f4e44623`	36.64	1.312	1.864	1.421 [1.310, 1.541]	9.46	13.45
`icv1-f2m41-tm2308219-7f48b14a`	39.00	1.966	3.185	1.489 [1.433, 1.547]	16.19	24.10
`icv1-f2m53-tm56619371-dac20a85`	44.26	2.728	3.845	1.409 [1.382, 1.438]	23.99	33.81

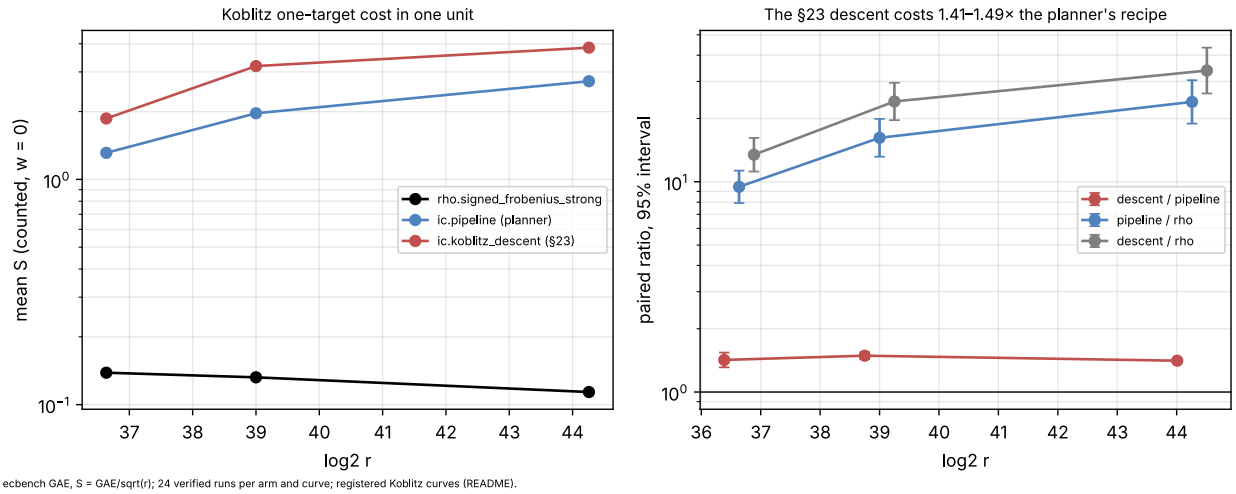


Figure 1: Counted S per arm, and the paired ratios.

The planner's recipe is the cheaper one: the descent costs 1.41–1.49× it at every size, with every interval above 1, so P1 fails and the routing stays. Rho is 9.5–34× cheaper than either.

Table A's 4.7–11.8× was mixed-unit: rho there is priced in canonical step units, while counted rho folds by 2n. The descent's cold cost is mostly reusable precomputation. A many-target comparison needs batch rho and is a different question.

The band cannot price the descent above w = 0 until four of its native categories are counted. Engineering class; no leaderboard row moves. Deviation: no resume checkpoint could be uploaded (no token); the owner said to proceed, and all inputs and outputs are pushed.